How-To Library

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Optimizing the Data Integration Service to Process Concurrent Web Services

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Abstract

The Data Integration Service runs concurrent web service requests according to the properties that you configure on the Data Integration Service and the application properties that you configure for each web service object. When you optimize the properties that affect web service concurrency, you can improve performance.

Supported Versions

- Informatica 10.2.2

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Overview

You can update the properties on the Data Integration Service to better manage and run concurrent web services. You can also update the application properties on the web service objects. When you optimize the properties that impact web service concurrency, you can increase performance.

Before you configure the Data Integration Service to run concurrent web services, consider the following factors:

- The number of concurrent request that you want to run
- The response time requirements
- The number of CPUs on the machine that runs the Data Integration Service
- The amount of memory available on the machine that runs the Data Integration Service

Tip: If the domain is set up for monitoring, verify that you use a separate Model repository to store the monitoring statistics. Performance may decrease if you use the same Model repository for Data Integration Service statistics and for run-time operations.

Updating the Data Integration Service properties

To optimize concurrent web service performance, review and optionally update the Data Integration Service properties.

You set the properties on the **Properties** tab and the **Processes** tab for the Data Integration Service in the Administrator tool.

Updating properties on the Properties tab

To find the Data Integration Service properties, select the Data Integration Service in the Navigator of the Administrator tool and click the **Properties** tab. The Data Integration Service properties display in the contents panel.

Review and optionally update the following properties:

DTM Keep Alive Time

Specifies the number of milliseconds that the DTM instance remains open after it completes a request. Web service requests that are issued against the same operation can reuse the open instance. Find the property under *Web Service Properties*. The default value is 5000.

Use the keep alive time to increase performance when the time required to process the request is small compared to the initialization time for the DTM instance. When the time elapses, the DTM process ceases to be active and the Data Integration Service starts a new DTM process for the next incoming request. If a request fails, the DTM instance terminates.

Set the *DTM Keep Alive Time* property to a value that is greater than the idle time that you anticipate between web service requests. If you set the value to zero, the Data Integration Service does not keep the DTM instance in memory.

You can also set a DTM Keep Alive Time on each web service that you deploy to the Data Integration Service. If you set the property on a deployed web service, the value for the deployed web service overrides the value that you set on the Data Integration Service. However, if you set the property on a deployed web service to -1, the web service uses the value that you set on the Data Integration Service.

Note: Using an open DTM process increases performance. However, keeping the DTM active consumes memory. Consider the impact on memory consumption when you configure the property.

Launch Job Options

Specifies whether the Data Integration Service runs all jobs in the same service process or in separate processes. By default, the Data Integration Service runs jobs in separate local processes. To optimize performance, set the property to run jobs in the service process.

Maximum On-Demand Execution Pool Size

Specifies the maximum number of on-demand jobs that can run concurrently. Job types include data previews, profiling jobs, REST and SQL queries, web service requests, and mappings. The Data Integration Service runs on-demand jobs immediately if enough resources are available. Find the property under *Execution Options*. The default value is 10.

Set a maximum on-demand pool size that can accommodate the maximum number of concurrent jobs that Developer tool clients will run on the Data Integration Service. Bear in mind that users can also deploy SQL Data Services requests and web requests, which can also draw on the on-demand execution pool size. The maximum number of concurrent jobs that a given Developer tool client can run is 10. All jobs that the Data Integration Service receives draw on the execution pool.

When you increase the execution pool size, the DTM process of the Data Integration Service consumes more resources, including CPU, memory, and system I/O. Set an execution pool value that takes account both of the resources available on the machine and the data service requirements.

Note: The Maximum On-Demand Execution Pool Size value that you set applies to a single Data Integration Service process. When the Data Integration Service runs on a grid, the maximum number of deployed and on-demand jobs that can run concurrently across the grid is calculated as follows:

*Maximum on-demand pool size * Number of running service processes*

Tip: Set the on-demand pool size according to the computer resources. Next, set the maximum concurrent requests on the **Processes** tab. Then, set the maximum backlog requests on the **Processes** tab.

Updating properties on the Processes tab

To find the Data Integration Service process properties, select the Data Integration Service in the Navigator of the Administrator tool and click the Processes tab. The Data Integration Service process properties display in the contents panel.

Review and optionally update the following properties:

Maximum Concurrent Requests

Specifies the maximum number of HTTP or HTTPS connections that can be made to the Data Integration Service process. The default value is 200. Find the property under *HTTP Configuration Properties*.

The property determines the number of requests that the Data Integration Services accepts before it sends the requests to the Data Integration Service backlog. Set the property to a value that is lower than the **Maximum On-Demand Execution Pool Size** value on the **Properties** tab. Restart the Data Integration Service to apply your changes.

Note: You can also set a Maximum Concurrent Requests value on each web service. The sum of the maximum concurrent requests and maximum backlog requests that you set on the Data Integration Service must be greater than or equal to the maximum number of concurrent requests across all web services.

Maximum Backlog Requests

The maximum number of HTTP or HTTPS connections that can wait in a queue for the current Data Integration Service process. The default value is 100. Find the property under *HTTP Configuration Properties*.

To prevent the Data Integration Service from rejecting a request because the backlog is full, ensure that the backlog value is greater than or equal to the number of concurrent requests across all web service applications minus the number of concurrent requests that the Data Integration Service specifies.

Calculating the backlog size

Let's say that you deploy five web service applications to the Data Integration Service, each with Maximum Concurrent Requests value of 100. You set the Maximum Concurrent Requests property value on the Data Integration Service to 50.

You can calculate the required backlog capacity in the following way:

Total number of concurrent requests available across all web service applications - the maximum concurrent requests available on the Data Integration Service

The required backlog size is (100 + 100 + 100 + 100 + 100) - 50 = 450. Set a value of 450 on the Maximum Backlog Requests property.

Maximum Heap Size

Specifies the amount of RAM to allocate to the Java Virtual Machine (JVM) that runs the Data Integration Service. The default value is 640 megabytes. Find the property under *Advanced Properties*.

Consider increasing the maximum heap size when the Data Integration Service must handle a large number of concurrent requests.

When you set the value, use one of the following letters to specify the units of measurement:

- b for bytes
- k for kilobytes
- m for megabytes
- g for gigabytes

Updating the web service application properties

To optimize concurrent web service processing, consider updating the application properties for each web service.

To find the application properties, select the Data Integration Service in the Navigator of the Administrator tool, click the **Applications** view, and select the web service.

Review and optionally update the following properties:

Trace Level

Determines the types of message that the Data Integration Service can write to the run-time web service logs. The Data Integration Service can write messages at the following levels of severity: DEBUG, TRACE, INFO, WARNING, ERROR, and FATAL. The least severe message type is DEBUG. The most severe message type is FATAL. The default type is INFO.

The number of web service log files that the Data Integration Service writes and maintains can affect performance. To turn off run-time web service logging for the Data Integration Service, set the Trace Level property to OFF.

Alternatively, if you want the Data Integration Service to generate run-time web service logs in cases of error, set the **Skip Log Files** property of the Data Integration Service to true and set the value of the property to INFO or higher. Find the **Skip Log Files** property under *Web Service Properties* on the **Properties** tab.

Maximum Concurrent Requests

Specifies the maximum number of requests that the selected web service can process at one time. The default value is 10.

Set the property to a value that is less than or equal to the *Maximum Backlog Requests* value on the Data Integration Service. The number can exceed the *Maximum Concurrent Requests* value on the Data Integration Service.

Web service processing path

A web service request takes the following path:

- A web service client sends a SOAP request to the Data Integration Service. If the Data Integration Service can accept the request without exceeding the specified maximum number of concurrent requests, the Data Integration Service adds the request to the list of active web service requests. Otherwise, the Data Integration Service tries to add the request to the backlog.
- The Data Integration Service can add a request to the backlog when the number of request in the backlog is lower than the specified maximum number of backlog requests. Otherwise, the Data Integration Service rejects the request. Requests in the backlog are added to the active queue as space becomes available.
- After the request enters the Web Service active queue, the Web Service module tries to add the request to the DTM active queue so that the DTM can process the request. If the DTM active queue can accept the request without exceeding the specified maximum execution pool size, the DTM runs the request and returns the SOAP response to the web service client. Otherwise, the SOAP request fails.

Example: concurrent request limits

You can configure a maximum number of concurrent requests for the Data Integration Service process and for the web service object. Web service requests fail when the number of concurrent web service requests exceed the maximum value that you configure for the web service.

With the following configuration, the Data Integration Service process can accept 200 concurrent requests and the web service can accept 10 concurrent requests:

Property Type	Property Name	Property Value
Data Integration Service Processes	Maximum Concurrent Requests	200
Applications: [web service name]	Maximum Concurrent Requests	10

As a result, web service requests fail when the Data Integration Service process receives more than 10 concurrent requests for the web service.

Tip: Set the web service maximum value to a value that reflects the number of concurrent web service requests that you expect in your environment.

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